

fn find_min_covering

Michael Shoemate (@Shoeboxam)

This proof resides in “**contrib**” because it has not completed the vetting process.

Proves soundness of `find_min_covering` in `mod.rs` at commit `f5bb719` (outdated¹).
`find_min_covering` attempts to return the smallest covering from `sets` that spans `must_cover`.

1 Hoare Triple

Precondition

Compiler-verified

- Argument `must_cover` is of type `BTreeSet<T>`
- Argument `sets` is of type `HashMap<BTreeSet<T>, u32>`
- Generic `T` is some type that implements `Hash` and has total `Ord`, so that it can be used in a B-tree set.

Caller-verified

None

Pseudocode

```
1 def find_min_covering(  
2     must_cover: set[T], sets: list[set[T], u32]  
3 ) -> list[tuple[set[T], u32]] | None:  
4  
5     covered = list() #  
6  
7     while must_cover: #  
8  
9         def score(pair):  
10             by, weight = pair  
11             return len(by & must_cover), -len(by), -weight  
12  
13         best_match = max(sets.items(), key=score)  
14  
15         if best_match is None or best_match[0].isdisjoint(must_cover):  
16             return None  
17         best_set, weight = best_match  
18  
19         must_cover -= best_set #  
20         covered[best_set] = weight  
21  
22     return covered
```

¹See new changes with `git diff f5bb719..378303f rust/src/domains/polars/frame/mod.rs`

Postcondition

Theorem 1.1. Return a subset of **sets** whose intersection contains **must_cover**, or None.

Proof. All that needs to be proven is that the return set covers **must_cover**. While the algorithm makes a best effort to minimize the cardinality of the cover, nothing about optimality of the algorithm has been proven.

The algorithm initializes with an empty cover on [5](#) and continues to run until all elements have been added to the cover (see [7](#)). If there are no remaining sets that intersect with **must_cover**, then the algorithm terminates without a cover, which is a valid output.

Otherwise, on **state**, a new set is added to the cover and those elements in **must_cover** are removed. The algorithm only terminates once all members of **must_cover** have had sets that include them added to **covered**.

Therefore **covered** returns a subset of **sets** whose intersection contains **must_cover**, or None.

□